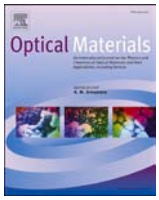
Optical Materials 149 (2024) 114979



Contents lists available at ScienceDirect

Optical Materials

journal homepage: www.elsevier.com/locate/optmat



Research Article



The wet-etching behaviors of the SiO_2 layer in $LiNbO_3$-on-insulator structure

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ARTICLE INFO

Keywords:
Lithium niobate thin film
LNOI
Wet etching
HF etching

ABSTRACT

The wet-etching behaviors of the SiO_2 layer in $LiNbO_3$-on-insulator structure were studied by the concentration of 3.6 % HF acid at different times. Drill etching occurs on the upper surface of the SiO_2 layer at a rate of 0.4278 μm/min. The transverse etching rate of the SiO_2 layer is 0.3321 μm/min, which is slightly less than the drill etching rate. So a triangular etched region is formed in the HF etched SiO_2 layer at an angle of about 30°. Moreover, the etching morphology and rate are the same on both sides of the strip. We also studied the etching rate of micro - and nano-structures with HF acid is about 0.22 μm/min. We numerically calculated the comparison of mode sizes between partially removed SiO_2 layer and non removed SiO_2 layer strip waveguides.

1. Introduction

Lithium niobate ($LiNbO_3$) is an important optical material owning wide light transparent window, excellent nonlinear effect and acousto-optical properties [1]. In recent years, the single-crystal thin film lithium niobate on insulator (LNOI) has been developed rapidly, which retains basically or very close to the properties of the bulk material [2,3]. The emergence of LNOI structure overcomes the excessive size of the devices based on $LiNbO_3$ bulk materials, the limited of the nonlinear efficiency of the optical devices, the optical gain and the compactness of the optical path. Single-crystal $LiNbO_3$ thin films provide a new integrated optical platform to give $LiNbO_3$ new optical and acoustic properties. Low-loss optical waveguides [4], high-performance electro-optical modulators [5–7], high-efficiency nonlinear optical devices [8–10] and on-chip integrated lasers [11] have been realized on this material platform, which has attracted great attention in the optical field. The microfabrication technology and equipment of processing semiconductor materials for integrated circuits chips is very mature, which also can be used to process LNOI materials. The methods of fabricating various devices on LNOI wafers including electron-beam lithography combined an optimized Ar -based dry etching to form ridge waveguides for realizing ultrahigh-efficiency wavelength conversion [12], diamond blade dicing to form low loss ridge waveguide [13], plasma reactive ion etching to achieve low loss rib waveguides for realizing efficient edge coupling [14] and the combination of femtosecond-laser ablation, focused ion beam milling and selectively etched by HF acid for broadband quasi-phase-matched harmonic generation [15] have been reported already. The study of etching technology plays an important role in the preparation of various devices. Whether dry etching or wet etching, they have their own advantages in etching rate, fineness and controllability. Wet etching is a low cost and high efficiency pattern transfer method at present. In the LNOI structure, using wet etching to remove SiO_2 has the following advantages: firstly, it can increase the refractive index difference, thereby strengthening the limitation of light and reducing the device volume; Secondly, it can reduce the loss generated at the interface between $LiNbO_3$ and SiO_2 due to the presence of SiO_2; Thirdly, removing SiO_2 can result in a higher extinction ratio for structures similar to photonic crystals. Therefore, it is of great significance to study the wet etching of SiO_2 layers in the LNOI structure. As far as we know, to date, no one has conducted research on such aspect in LNOI structure.

In this paper, we studied the wet etching of the SiO_2 layer in LNOI by a diluted HF solution including etching rate, etching angle and so on. In addition, we also calculated the optical guiding characteristics of strip waveguides that can be prepared by such wet etching method.

2. Experiments and details

The LNOI wafer with a 700 nm single-crystal $LiNbO_3$ thin film was fabricated by the smart cut technology in NANOLN, Jinan Jingzheng

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https://doi.org/10.1016/j.optmat.2024.114979
Received 25 August 2023; Received in revised form 24 December 2023; Accepted 25 January 2024
Available online 1 February 2024
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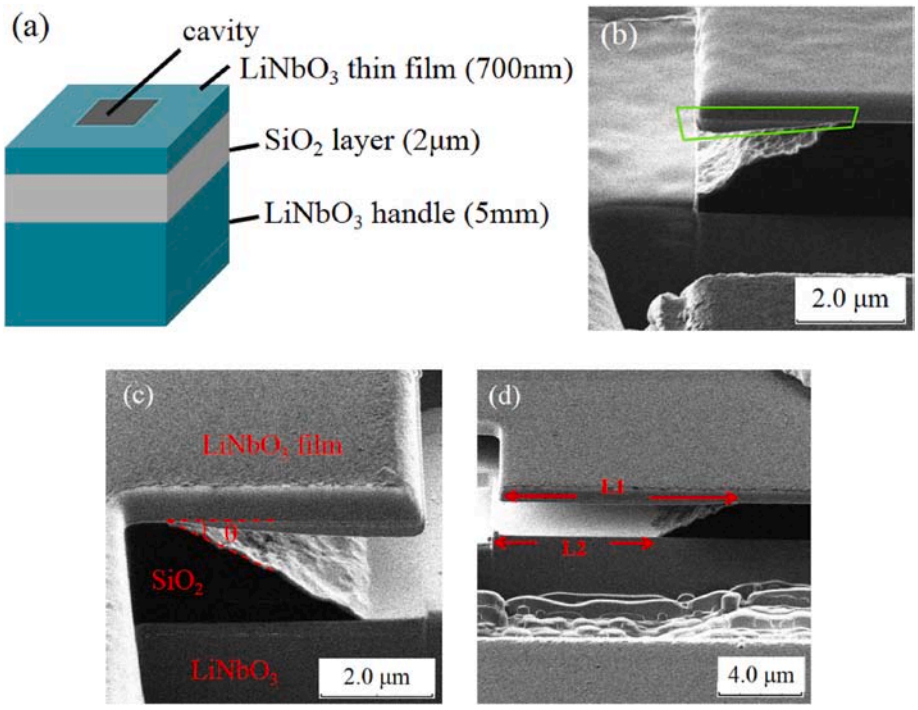


Fig. 1. (a) The schematic diagram of the cavity structures on the LNOI wafer; The images of the cross sections with a 60° tilt (split by FIB) after HF etching at different times: (b) 6 min, (c) 14 min, and (d) 30 min, respectively.

Table 1
The etching time and length of SiO_2 layer by 3.6 % HF acid at 30 °C.

Time (min)	3	6	7	10	14	21	30	35
L1 (μm)	0.67	2.30	2.07	3.62	5.25	7.30	11.76	15.00
L2 (μm)	–	–	1.00	1.40	3.18	4.50	8.00	11.00

Electronics Co., Ltd. The $LiNbO_3$ thin film is bonded on a SiO_2 buffer layer with 2 μm thickness. The SiO_2 layer is deposited by plasma enhanced chemical vapour deposition on a $LiNbO_3$ handle. Firstly, the cavity structures on the LNOI wafer with different depths such as just through the $LiNbO_3$ film (about 700 nm deep), through the SiO_2 layer (greater than 2.7 μm deep), or to a certain location in the SiO_2 layer (greater than 700 nm but less than 2.7 μm), were prepared by focused ion beam (FIB). The schematic diagram is shown in Fig. 1(a). Secondly, samples were immersed in the 3.6 % HF acid at 30 °C with different times from 3 to 35 min. This concentration of HF can quickly etch away the SiO_2 layer but does not damage the $LiNbO_3$ film [16].

3. Results and discussion

To observe the HF etching phenomena, the samples were split by FIB and the cross sections with a 60° tilt were shown in Fig. 1. The structures in Fig. 1 (b), (c) and (d) were formed after HF etching for 6, 14, and 30 min, respectively. The long narrow triangle framed by a green polygon at the lower edge of the lithium niobate film end face may be caused by the trembling of the film during the FIB cutting process, in Fig. 1 (b). A triangular etched region is formed in the HF etched SiO_2 layer. It shows that the etching rate of the upper surface of SiO_2 layer (that is, the side near the $LiNBO_3$ film) is relatively fast. This phenomenon can be interpreted as the occurrence of drill etching in wet etching on the upper surface of the SiO_2 layer. It is measured that the tilt angle θ is about 30°, and the angle remains same as the etching time increased as showed in Fig. 1(b), (c) and (d).

We used the least square method to fit the etching rate of HF acid on the upper surface of the SiO_2 layer according to the data in Table 1, as shown in Fig. 2 (a). The horizontal coordinate represents the etching time, and the vertical coordinate represents the etching length of the upper surface of the SiO_2 (L1). R is the correlation coefficient, and the closer the $|R|$ value is to 1, the denser the distribution of experimental data points near the fitted line. It is appropriate to use a linear function for regression. The square of the correlation coefficient R is equal to 0.9896, indicating that there is a close linear relationship between the length and the etching time. From the slope in the formula, we can see

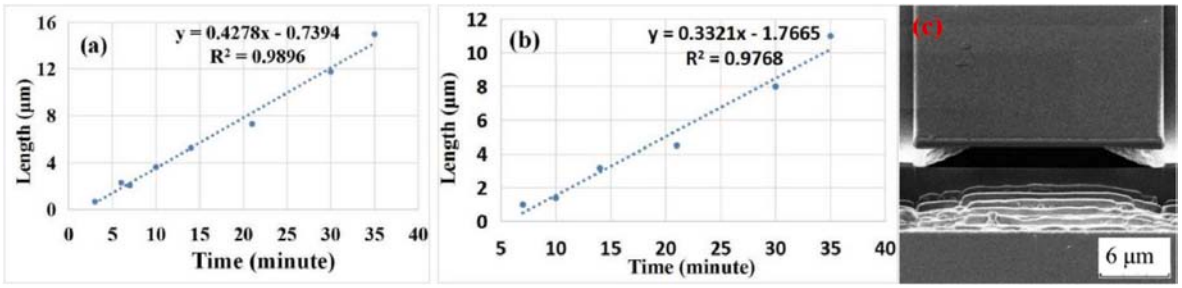


Fig. 2. (a) The drill etching rate of HF acid on the upper surface of the SiO_2 layer; (b) The transverse etching rate of HF acid about the SiO_2 layer; (c) A strip etched with HF acid after 14 min.

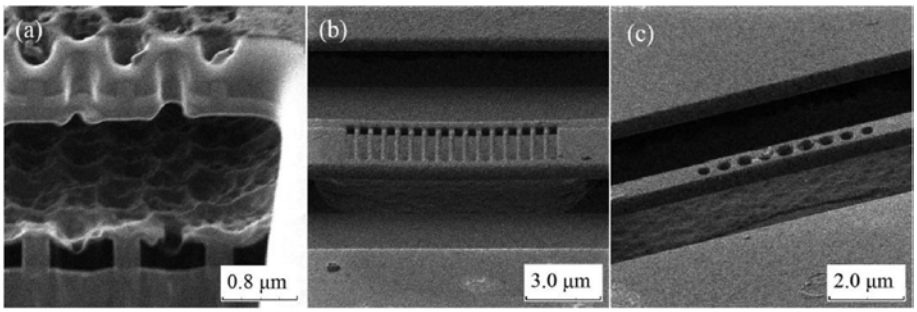


Fig. 3(a). The image of a triangular array structure etched with HF acid for 7 min; (b) and (c) the one-dimensional air-bridge structures prepared by the combination of FIB etching and HF etching for 10 min.

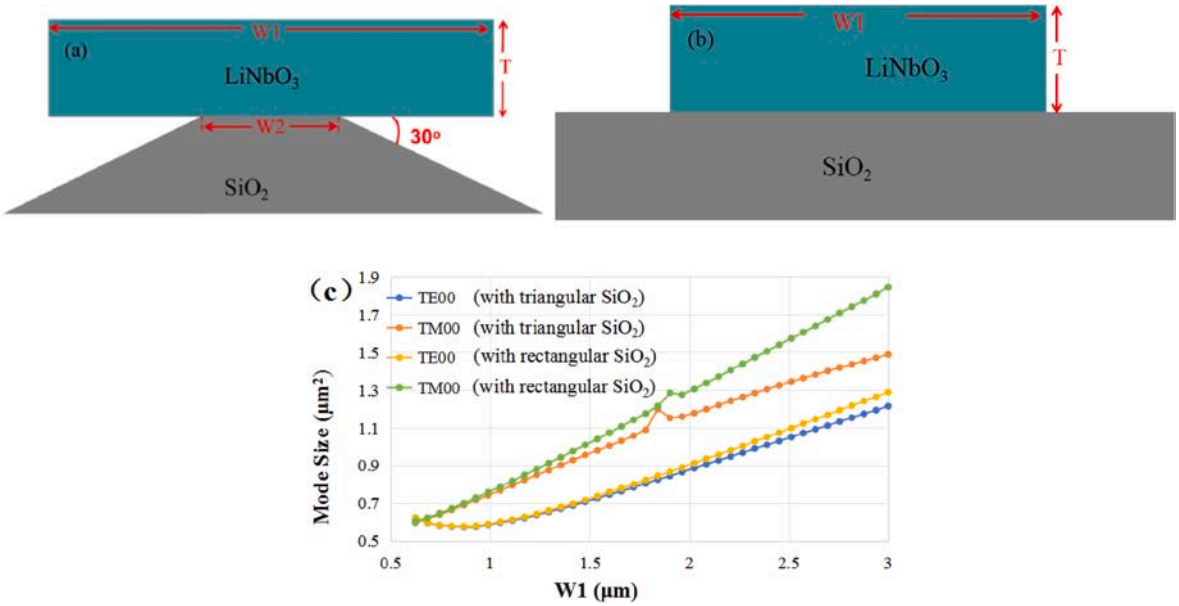


Fig. 4. (a) A schematic diagram of cross section for the strip waveguide with the triangular SiO_2 layer; (b) A schematic diagram of cross section for the strip waveguide with the rectangular SiO_2 layer; (c) The relationship between the mode size of the strip waveguide in figures (a) and (b) and W1.

that the drill etching rate is about 0.4278 μm/min. In order to get the transverse etching rate of HF acid solution, the SiO_2 layer was etched through longitudinally when the structure is prepared with FIB. We also used the least square method to fit the transverse etching rate of HF acid about the SiO_2 layer as shown in Fig. 2 (b). The horizontal coordinate represents the etching time, and the vertical coordinate represents the length of the lower surface of the silica layer (L2) being etched. The square of the correlation coefficient of 0.9768 indicates a roughly constant transverse etching rate of 0.3321 μm/min. It is found that the drill etching rate on the upper surface of SiO_2 is slightly faster than the transverse etching rate on the lower surface, which is why the triangular area shown in Fig. 1 is formed. This phenomenon has been reported in other literature [15]. We think that there are two main reasons for the different etching rates. Firstly, the bonding force (formed during the bonding process) between SiO_2 layer and $LiNbO_3$ thin film is smaller than that between SiO_2 layer and $LiNbO_3$ handle (formed during thermal growth); Secondly, because HF acid flows down from above, after consumption, the concentration of HF acid at the upper interface is greater than that at the lower interface. Fig. 2 (c) shows a strip etched with HF acid after 14 min. From the structure, it can be found that the triangular area and the etching length of the left and right sides of the strip structure are symmetrical, indicating that the etching is the same on both sides.

We also studied the etching of micro - and nano-structures with HF acid. Fig. 3 (a) shows a triangular array structure with a period of 560 nm, a radius of 130 nm and a depth of 800 nm by FIB etching, which is etched with HF acid for 7 min. It can be estimated that the etching rate is about 0.22 μm/min, which is less than the transverse etching rate and the drill etching rate. After reasonable design, we can prepare a variety of different micro and nano size air-bridge structures. Fig. 3 (b) and (c) show one-dimensional air-bridge structures prepared by the combination of FIB etching and HF etching for 10 min. In Fig. 3(c), the impurity on the fourth hole is an uneven gold film, which is used to conduct electricity during the observation process. We have also used this method to prepare the air-bridge photonic crystal structures, and the relevant content has been published [17].

We have numerically calculated the optical characteristics of the strip waveguide that can be prepared by the combination of FIB (or other dry etching methods) and HF etching like in Fig. 2(c). The Full-Vector Finite Difference method was used to analyse the parameters of the waveguide modes. W2 represents the width of the upper surface of SiO_2, W1 and T represent the width and thickness of the *x*-cut $LiNbO_3$ thin film, respectively, shown in Fig. 4(a). In order to obtain the optical confinement of the two strip waveguide structures shown in Fig. 4(a) and (b), we numerically calculated the relationship between the mode size and W1 (from 0.6235 μm to 3 μm) for the modes TE00 and TM00 in Fig. 4(c). The mode size is the product of 1/e of the horizontal light intensity distribution and 1/e of the vertical light intensity distribution for the simulated mode. Parameter T is fixed at 560 nm and W2 at 0.6235 μm. It can be seen that the mode size of the strip waveguide

shown in Fig. 4(a) is smaller than that of the strip waveguide shown in Fig. 4(b), regardless of whether it is TE00 or TM00. The smaller the mode, the stronger the limitation on light. Therefore, the study on wet etching of SiO_2 layers in LNOI structure is beneficial to the development of integrated optical devices to a certain extent.

4. Conclusions

In conclusions, the diluted HF (3.6 %) etching behaviors of LNOI structures were studied. The least square method was used to fit the etching rate, and the calculated correlation coefficient showed that there was a close linear relationship between etching length and time. The drill etching occurred on the upper surface of SiO_2, the etching rate is about 0.4278 μm/min. The transverse etching rate of the SiO_2 layer is 0.3321 μm/min. It is because of the different etching rates that a triangle area with an angle of about 30° is formed in the etched SiO_2 area. Theoretically, it has been calculated that waveguides prepared on the basis of wet etching SiO_2 will have small mode size. Next, we will use the method of wet etching to partially remove SiO_2 to prepare waveguide structures and other optical structures based on this study. We estimate that the waveguide structure will have high-quality features such as low loss and small mode volume.

CRediT authorship contribution statement

Shao-Mei Zhang: Writing – original draft. **Jin-Hua Zhao:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

We thank professor Hui Hu from Shandong University for helpful discussion. We are grateful to the research center of Nanoln for providing the LNOI samples.This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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